AVR C Programming Lab Programs

Simple Logic Programs

Prepared for Practical Examination

Program 1

Question: Configure PORTC as Output and Display 0x55 Continuously

```
#include <avr/io.h>

int main()
{
    DDRC = 0xFF;

    while(1)
    {
        PORTC = 0x55;
    }
}
```

Simple Logic: Logic: Make PORTC output and continuously send 0x55.

Program 2

Question: Blink LED Connected to PC0 with 500 ms Delay

```
#include <avr/io.h>
#include <util/delay.h>

int main()
{
    DDRC = 0xFF;

    while(1)
    {
        PORTC = 0x01;
        _delay_ms(500);

        PORTC = 0x00;
        _delay_ms(500);
    }
}
```

Simple Logic: Logic: LED ON → Delay → LED OFF → Delay.

Program 3

Question: Read PB0 and Turn ON LED Connected to PC0

```
#include <avr/io.h>

int main()
{
    DDRB = 0x00;
    DDRC = 0xFF;

    while(1)
    {
        if(PINB == 0x01)
            PORTC = 0x01;
        else
            PORTC = 0x00;
    }
}
```

Simple Logic: Logic: Read switch and control LED.

Program 4

Question: Copy Data from PORTB to PORTC

```
#include <avr/io.h>

int main()
{
    DDRB = 0x00;
    DDRC = 0xFF;

    while(1)
    {
        PORTC = PINB;
    }
}
```

Simple Logic: Logic: Read PORTB and display on PORTC.

Program 5

Question: Convert Packed BCD (23) to ASCII

```
#include <avr/io.h>

int main()
{
    unsigned char bcd = 0x23;
    char ascii1, ascii2;

    ascii1 = (bcd / 16) + '0';
    ascii2 = (bcd % 16) + '0';

    while(1);
}
```

Simple Logic: Logic: Split BCD digits and convert to ASCII.

Program 6

Question: Convert Two ASCII Digits into One Packed BCD

```
#include <avr/io.h>

int main()
{
    char a = '2';
    char b = '3';

    unsigned char bcd;

    bcd = ((a - '0') * 16) + (b - '0');

    while(1);
}
```

Simple Logic: Logic: Remove ASCII value and combine digits.

Program 7

Question: Blink LED Connected to PC0 Using Timer0

```
#include <avr/io.h>

int main()
{
    DDRC = 0xFF;

    TCCR0 = 0x05;

    while(1)
    {
        if(TIFR == 0x01)
        {
            PORTC = ~PORTC;
            TIFR = 0x01;
        }
    }
}
```

Simple Logic: Logic: Wait for Timer0 overflow and toggle LED.

Program 8

Question: 8-bit Event Counter Using Timer0

```
#include <avr/io.h>

int main()
{
    DDRB = 0x00;
    DDRC = 0xFF;

    TCCR0 = 0x06;
    TCNT0 = 0x00;

    while(1)
    {
        PORTC = TCNT0;
    }
}
```

Simple Logic: Logic: Count external pulses and display count.

Important AVR Registers

DDRx - Sets input/output direction

PORTx - Writes output

PINx - Reads input

TCCR0 - Timer0 control

TCNT0 - Timer0 counter

TIFR - Timer flags

TOV0 - Timer overflow flag